Math 152 Problem Set 6

Due 11/13/2026, at 10:00 PM

Problem 1. Consider the plane region below the graph of $\exp(-x^2)$ over the interval $[0, 1]$. Compute the volume of the solid obtained by revolving this region around the y axis.

Problem 2. Let f and g be differentiable functions. The function

$$h(x) = f(x)^{g(x)}$$

makes sense where $f(x) > 0$. Derive a formula for the derivative of h in terms of f, g, f', and g'.

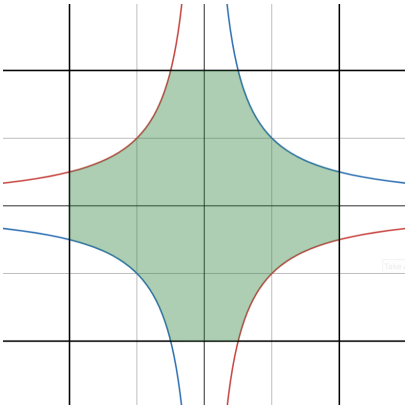
Problem 3. Find the area of the region depicted below, bounded by the following curves:

$$y = 1/x$$

$$y = -1/x$$

$$x = \pm 2$$

$$y = \pm 2$$



Problem 4. Suppose that f and g are differentiable functions on an interval $[a, b]$, such that f and g have no zeroes, and their logarithmic derivatives are equal:

$$\frac{f'(x)}{f(x)} = \frac{g'(x)}{g(x)}$$

for all $x \in [a, b]$. Argue that $g(x) = C \cdot f(x)$ for some nonzero constant C[1].

Problem 5. Evaluate the following integral:

$$\int_1^2 \left[\frac{2^{x+1}}{1 + 2^x} + (x - 1)^{\sqrt{2}} \right] \mathrm{d}x$$

[1] At least two approaches are possible here. One is to write the logarithmic derivatives in as derivatives of natural logarithms of absolute values, and compare these. Another is to consider the function $g(x)/f(x)$. (What is its derivative?)

Problem 6 (Optional)**.** In this problem we will derive some estimates on the growth rate of the exponential function.

(a) Explain why

$$\exp(x) = 1 + \int_0^x \exp(t)\mathrm{d}t$$

(b) Argue that $\exp(x) \geq 1$ for $x \geq 0$.[2]

(c) Use parts (a) and (b) to show that $\exp(x) \geq 1 + x$ for $x \geq 1$, and apply the same technique again to show that $\exp(x) \geq 1 + x + x^2/2$.

(d) Assume that you have the estimate

$$\exp(x) \geq 1 + x + \frac{x^2}{2} + \cdots + \frac{x^n}{n!}$$

for $x \geq 0$, where $n! = 1 \cdot 2 \cdot \cdots \cdot n$ is the factorial function. Use (a) to improve this estimate to

$$\exp(x) \geq 1 + x + \frac{x^2}{2} + \cdots + \frac{x^n}{n!} + \frac{x^{n+1}}{(n+1)!}$$

This kind of argument, mathematical induction, allows us to conclude that the estimate from part (d) holds for all n. This means in particular that $\exp(x)$ eventually grows faster than any polynomial.

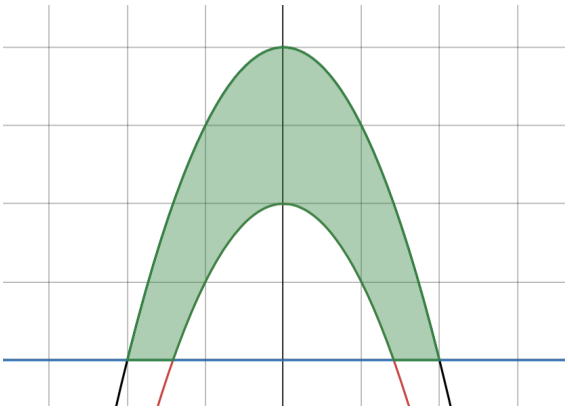
Review Problems

Problem. Compute the area of the plane region depicted below, bounded by the following curves:

$$y = 4 - x^2$$

$$y = 2 - x^2$$

$$y = 0$$



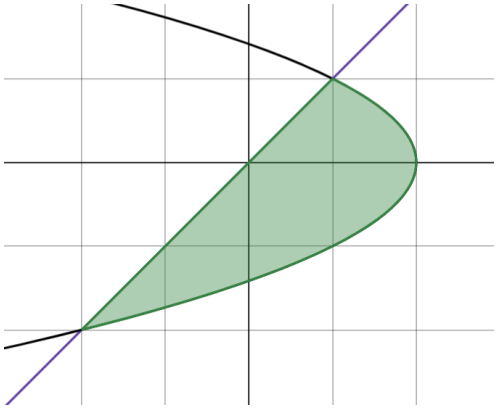
[2]Hint: Use the fact that $\exp(x) > 0$ everywhere.

Problem. Consider plane region below the graph of $x^{-1/2}$ over the interval $[1, 4]$. Compute the volume of the solid obtained by revolving this region about the x axis.

Problem. Compute the area of the plane region depicted below, bounded by the following curves:

$$y = x$$

$$x = 2 - y^2$$

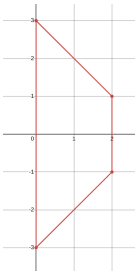


Problem. Compute the volume of the solid obtained by revolving the plane region between the curves $y = 2x^2$ and $y = x + 1$ over the interval $[0, 1]$ around the y-axis.

Problem. Evaluate the following integral.

$$\int_{-\pi/4}^{\pi/4} \frac{\cos(2x)}{2 + \sin(2x)} \mathrm{d}x$$

Problem. Find the volume of a solid whose base is the trapezoid depicted below, and whose cross sections perpendicular to the x-axis are triangles whose base and height are equal.



Problem. Find the average value of $2\sin(x) + \cos(x)$ on the interval $[3\pi, 9\pi/2]$.

Problem. Consider the function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3 + 3x + 1$.

(a) Explain why this function is injective.

(b) Explain why this function is surjective.

(c) Let g be the inverse of f. Using $f(-1) = -3$, compute $g'(-3)$.